Rui Li

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Summary

PhD candidate in Computer Science at Leiden University, specializing in Explainable AI for malware detection and analysis. My research evaluates explanations produced by malware detection models to improve model reliability and support security analysts' trust, covering local and global explainability as well as supervised and clustering-based detectors. Moreover, I investigate how Large Language Models (LLMs) can connect low-level features with high-level malicious behaviours, including semantic alignment between malware features and MITRE ATT&CK techniques. I also have experience working with large-scale data processing and machine learning (incl. LLM) pipelines.

Education

Oct 2021 – Oct 2026 **PhD in Computer Science**, *Leiden University*, Netherlands.
Research Focus: Explainable AI in Malware Detection.
Supervisor: Olga Gadyatskaya (o.gadyatskaya@liacs.leidenuniv.nl).

Sept 2017 – June 2020 **Master of Electronic Engineering**, *Beijing Institute of Technology*, China.

Sept 2013 – June 2017 **Bachelor of Electronic Engineering and Economics (Dual Degree)**, *Beijing Institute of Technology*, China.

Research Experience

Large Language Models (LLMs)

- Designed LLM-based semantic alignment pipelines for structured knowledge reasoning on large real-world datasets.
- Developed and compared prompting strategies (e.g., zero-shot, few-shot) to improve semantic matching performance.
- Benchmarked commercial API-based and open-source LLMs using ranking-based evaluation metrics (e.g., Accuracy, NDCG) to assess representation quality.
- *Related publication: SAC 2026*

Explainable AI and Model Interpretability

- Developed and evaluated interpretability methods for classification models and clustering models.
- Designed quantitative metrics (stability, robustness, effectiveness) to assess explanation quality and decision consistency.
- Conducted empirical studies comparing human-annotated explanations with model-generated explanations to evaluate explanation fidelity and model reliability.
- *Related publications: EuroS&PW 2024; NSS 2023*

Work Experience

July 2020 – July 2021 **Security Software Test Engineer**, *China National FinTech Evaluation Center*, Beijing.
- Conducted security evaluation for Electronic Social Insurance Card software using Fortify, SonarQube, and Burp Suite, aligned with national security standards and covering authentication, data protection, communication security, cryptographic algorithms, and interface security.
- Performed security testing on Samsung and Golden River V-Trust Trusted Execution Environment (TEE), strengthening practical experience in security-sensitive assessment contexts.

April 2019 – Sept 2019 **Machine Learning Engineer (Intern)**, *Tsinghua University*, Beijing.
- Developed a multi-factor quantitative investment strategy generation system based on a random herd optimization algorithm.
- Designed and evaluated optimized sales allocation strategies using inventory and sales data from a manufacturing company.

Publications

Rui Li, and Olga Gadyatskaya. "APIMatch-ATT&CK: Automatically Mapping Android APIs to MITRE ATT&CK", ACM/SIGAPP Symposium On Applied Computing (SAC), 2026.

Rui Li, and Olga Gadyatskaya. "Position: The explainability paradox – Challenges for XAI in malware detection and analysis." IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), Vienna, Austria, 2024.

Rui Li, and Olga Gadyatskaya. "Evaluating rule-based global XAI malware detection methods." International Conference on Network and System Security (NSS). Cham: Springer Nature Switzerland, Aug 2023.

Rui Li, Senlin Luo, Limin Pan, Jingwei Hao, Hanqing Zhang, Senlin Luo, and Qian Wu. "A Highly Accurate Individual Network Security Awareness Evaluation and Group Index Calculation Method." Journal of Beijing Institute of Technology, June 2019.

Rui Li, Feiyang Sun, and Olga Gadyatskaya. "Evaluating Explainable Clustering Methods for Android Malware Family Attribution." (under review)

Rui Li, Shihang Yu, Oussama Soulimani, and Olga Gadyatskaya. "Multi-Metric Evaluation of Local and Global Explanations for Android Malware Detection." (under review)

Student Supervision Experience

2021–2022 **Oussama Soulimani**, *Bachelor thesis*.
"Evaluating Android Malware Detection Explanation".

2022–2023 **Feiyang Sun**, *Master thesis*.
"Explanation of XAI clustering methods on Android malware family categorization".

2024–2025 **Shihang Yu**, *Master thesis*.
"Explainable AI in Android Malware Detection".

2025–2026 **Priya Georgina Anuradhakumari Smit**, *Bachelor thesis*.
"Detecting Android Malware Using Graph Neural Networks".

Talks

2023 **Leiden University LIACS PhD seminar**, *Netherlands*.
"Evaluating Rule-based Global XAI Malware Detection Methods".

2023 **ICTopen**, *Netherlands*.
"Domain-specific properties for global XAI malware detection methods".

2024 **CompSys**, *Netherlands*.
"Explanation of Clustering Methods for Distinguishing Android Malware Family".

Service

2025 Program Committee Member, 3rd International Conference on Data Science & Artificial Intelligence (DSAI).

2025 Event Organization Support, Ethics in Cybersecurity and AI, Leiden University.

2025 Event Organization Support, LIACS PhD Weekend, Leiden University.

Awards

2021–2025 **Chinese Scholarship Council**, *China*.

Sept 2018 **Winning Prize, Global (Nanjing) AI application competition**, *China*.

June 2017 **Outstanding Graduates, Beijing Institute of Technology**, *China*.

Nov 2016 **Silver Award, International Genetically Engineered Machine Competition**, *USA*.

Mar 2016 **Meritorious Winner, Mathematical Contest In Modeling**, *USA*.

2013–2020 **University Scholarship (7 times)**, *China*.

Skills

Programming Python, R

Libraries/Tools PyTorch/TensorFlow, HuggingFace, Scikit-learn, Git, Linux

Languages English (full professional proficiency), Chinese (mother tongue), Dutch (A2)

Activities

2025 SIKS-course: Explainable AI, Netherlands Research School for Information and Knowledge Systems, Netherlands.

2024 Summer School, Foundations of Security Analysis and Design, Italy.

2024 Summer School, Security and Privacy in the Age of AI, KU Leuven, Belgium.

2019 Summer School, Machine Learning, Israel Institute of Technology, Israel.

2018 Youth Volunteer, China League Two.

2017 Exchange Student, Tour University, France.

2015–2017 Director of External Liaison Department, Student Union, School of Electronic Engineering, Beijing Institute of Technology.

2013–2017 Youth Volunteer Service Team, Beijing Institute of Technology.

2016 Youth Volunteer, Tsinghua-Santander World Challenges of 21st Century Program.